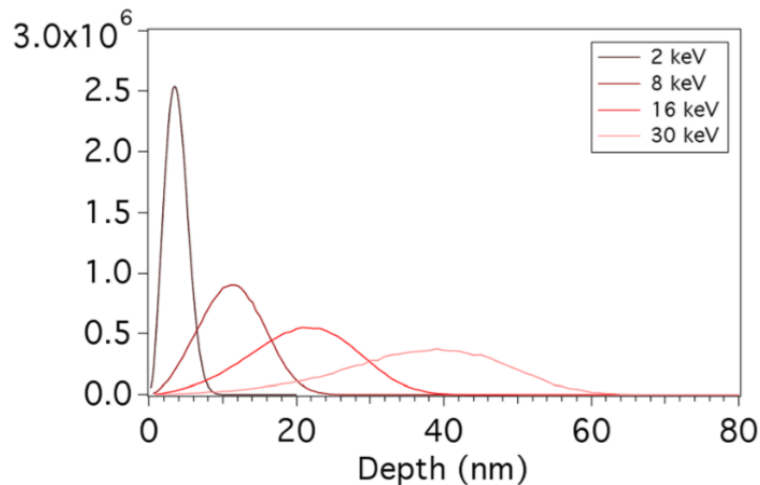


Tutorial 3

BACKGROUND

A scanned oxygen (O^+) focused ion beam (FIB) system was used to irradiate 4 regions of a single crystal diamond sample, using ion beam energies of 2, 8, 16 and 30 keV. The software SRIM was used to simulate the resulting depth profiles of implanted oxygen atoms (ie: plots of the number of oxygen atoms as a function of depth below the sample surface):

**PROBLEMS**

- A. The damage in the diamond crystal lattice consists of defects that include carbon vacancies and carbon interstitials generated by the oxygen ions. Do you expect the depth profiles of these vacancies and interstitials to overlap exactly with the above depth profiles of implanted O atoms? Why?

To answer this question, use software package SRIM to reproduce the above O depth profiles and to simulate the corresponding vacancy depth profiles. Include plots of both in your answer.

- B. In a second experiment, a Ga^+ FIB system was used to perform the same experiment (i.e., a Ga^+ beam with ion beam energies of 2, 8, 16 and 30 keV was used to irradiate 4 regions of single crystal diamond). Explain how the resulting depth profiles of implanted Ga atoms differ from the above depth profiles of implanted O atoms.